

Notes 9

Grep

- **Usage**
 - Grep is used to search text in a given file(s). Grep works line by line.
- **Formula**
 - `grep + option + search criteria + file(s)`
- **Examples**
 - Search any line that contains the word "dracula" in the given file.
 - `grep + 'dracula' + ~/Documents/dracula.txt`
 - Search any line that contains the word 'dracula' regardless of the case.
 - `grep + -i + 'dracula' + ~/Documents/Books/dracula.txt`
 - Display how many lines contain the matched string.
 - `grep + -c + 'dracula' + ~/Documents/Books/dracula.txt`

awk

- **Usage**
 - Awk is a scripting language used for processing and displaying text.
- **Formula**
 - `awk + option + {awk command} + file + file to save (optional)`
- **Examples**
 - Print the first column of every line of a file.
 - `awk + '{print $1}' + ~/Documents/Cvs/cars.csv`
 - Print first field of /etc/passwd file.
 - `awk + -F: + '{print $1}' + /etc/passwd`
 - Print the last field of the /etc/passwd file.
 - `awk + -F: + '{print $NF}' + /etc/passwd`

sed

- **Usage**
 - SED is a stream editor that performs operations on files and standard output.
- **Formula**
 - `sed + options + sed script + file`
- **Examples**
 - Replacing only the fourth occurrence per line in a file.
 - `sed + 's/false/true/4' + joke.json`
 - Replacing string on a specific line number.
 - `sed + '3 s/false/true/' + joke.json`
 - Replacing string on a range of lines.
 - `sed + '1,3 s/false/true/' + joke.json`

1. Explain how to use the pipe (|) for redirecting the output of a command to another. Include at least 3 examples

It sends the output of one command directly into another command as input.

- **Examples**

- Use grep to look for a string in a particular man page.
 - `man + ls + | grep + "Human-readable"`
- Display only the 2nd line in a file
 - `head + -2 + file.lst + | + tail + -1`
- Display only the options of the of any command from its man page.
 - `man + ls + | + grep + "^[:space:]*[:punct:]"`

2. Explain how to save the output of a command to a file (>). Include at least 3 examples

To save the output of a command to a file, you use the > redirection operator in a shell.

- **Examples**

- Save the current date to the file.
 - `date + > + date.txt`
- Save a directory listing.
 - `ls + -l + > + files.txt`
- Save command output including errors.
 - `ls + /nonexistent + > + output.txt + 2>&1`

3. Explain how to append the output of a command to a file. Include at least 3 examples

When we use > on a file that already exist and contains data, we overwrite whatever is inside the file.

- **Examples**

- Append a directory listing to a file.
 - `ls + -l + >> + files.txt`
- Append command output including errors.
 - `ls + /nonexistent + >> + errors.txt + 2>&1`
- Append the output of a script.
 - `./backup.sh + >> + backup.log`